

RESEARCH EXPERIENCE

- Generative Model and Multimodal Prediction** 2024 - Present
 - Video/Image Diffusion Model; World Model; Multimodal Large Language Model
- AI for Optics (Hardware-Software Interaction)** 2024 - Present
 - Meta-lens Imaging
- 3D Visual Perception** 2023 - 2024
 - Reconstruction; Surgical Simulation; Watermarking
- 2D Visual Perception** 2020 - 2024
 - Domain Adaptation; Open-Vocabulary/Set/World Object Detection; Noisy Labels
- Graph Learning** 2020 - 2024
 - Graph Matching; Causal Graph; Graph Motif
- Medical Imaging Analysis** 2020 - 2024
 - MRI Brain Analysis; Tooth Analysis; Medical Report Generation

WORK EXPERIENCE

- The Chinese University of Hong Kong (CUHK)** Mar. 2024 - Jan. 2025
 - Postdoc, Electronic Engineering
 - Supervisor:** Prof. Yixuan Yuan

EDUCATION

- City University of Hong Kong (CityU)** Sep. 2020 - Feb. 2024
 - Ph.D., Electrical Engineering
 - Supervisor:** Prof. Yixuan Yuan
 - 3.5-Year; Early Graduation
- Tianjin University (TJU)** Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA: 3.83/4.00 (Top 2%)

SELECTED PAPER ('[]' HIGHLIGHTS THE KEY WORD)

- [Diffusion] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
 European Conference on Computer Vision (ECCV), 2024, **Oral (1.8%)**
- [Video Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- [Open-World Learning] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2023, **Oral (1.8%)**
- [Open-World Learning] Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- [Domain Shift] SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023
- [Domain Shift] SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, **Best Paper Finalist (0.4%)**
- [Domain Shift] SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence (AAAI), 2022, **Oral (4.5%)**
- [AI for Optics] Stereo Vision Meta-Lens-Assisted Driving Vision**
 Xiaoyuan Liu, **Wuyang Li**, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
 ACS Photonics, 2024, **Selected as Cover Paper**

9. **[SAM] Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM**
Chenxin Li*, Yuzhi Huang*, **Wuyang Li**, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
Conference on Neural Information Processing Systems (NeurIPS), 2024
10. **[SAM] When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
Yifan Liu, **Wuyang Li**, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, **Early Accept (14.0%)**
11. **[Diffusion] Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
Xinyu Liu, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
12. **[Brain] FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
Zhibin He, **Wuyang Li**, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
13. **[3D] LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
Hengyu Liu*, Yifan Liu*, Chenxin Li*, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
14. **[Video Generation] Endora: Video Generation Models as Endoscopy Simulators**
Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, **Wuyang Li**, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
15. **[Brain] Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, **Wuyang Li**, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (TMI), 2024
16. **[Multi-Modality] MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
Qiushi Yang, **Wuyang Li**, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (ICCV), 2023
17. **[Noisy Labels] Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
Xinyu Liu, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, **Oral, Early Accept**
18. **[Noisy Labels] Towards Robust Adaptive Object Detection under Noisy Annotations**
Xinyu Liu, **Wuyang Li**, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
19. **[Object Detection] HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (TIP), 2021

AWARDS

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| <ul style="list-style-type: none"> • Outstanding Academic Performance Award, CityU • Outstanding research achievements • Research Tuition Scholarship, CityU • Outstanding research achievements • Ranked First in Polyp Detection Challenge • 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021) • National Scholarship, TJU • The highest undergraduate honor for a student in China • Silver Award in Tianjin Advanced Mathematical Competition • As a selected participant representing Tianjin University | <div>2022 - 2023</div> <div>2022 - 2023</div> <div>2021</div> <div>2017 - 2018</div> <div>2017</div> |
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SERVICES

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| <ul style="list-style-type: none"> • NeruIPS-24 Workshop Organizer [Link] • Organized the Advancements in Multi-modal Medical Foundation Models workshop [Proposal Link] • Research Funding Proposal • Assisted my advisor in completing several research funding proposals • Startup Funding [Link] • Successfully secured seed funding from HK Tech 300 and HKSTP • Founder of ScholaGO Education Technology Company Limited, Hong Kong [Link] • Founded a company to develop a multi-modal educational product • Conference and Journal Reviewer • ICLR-25; NeurIPS-24 (Top Reviewer); NeurIPS-24; CVPR-23/24; ICCV-23; T-PAMI; IJCV; TIP; TNNLS | <div>2024</div> <div>2022 - 2024</div> <div>2023</div> <div>2024</div> |
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